

```

%root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200GP', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
if exist('used_ds200GP', 'var')
    used_ds = used_ds200GP;
else
    used_ds = containers.Map()
    sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
    %
    sel_info.cuts = {cuts2fit200GP.cuts_off110f};
    sel_info.en_range = 30:10:120;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200off110_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir111')=sel_info;
    %
    sel_info.cuts = {cuts2fit200GP.cuts_off200f};
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200off200_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir111')=sel_info;
    %
    used_ds200GP = used_ds;
end

```

```

select= 'Ei200Br110dir111';
sel_info = used_ds(select);

```

```

cuts_list = sel_info.cuts

```

```

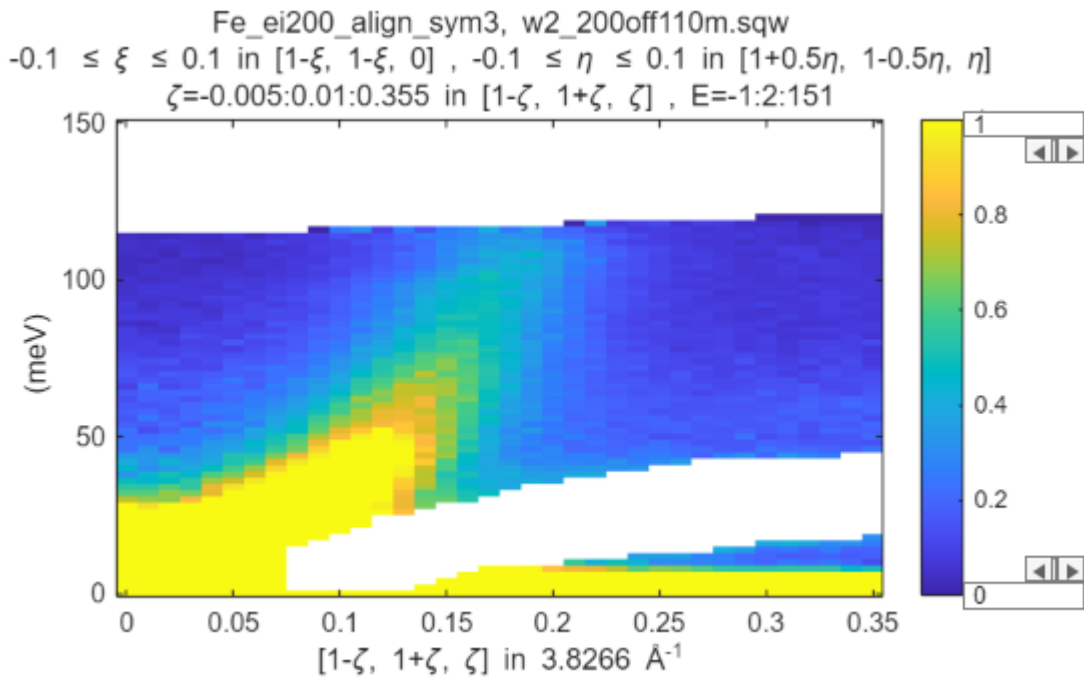
cuts_list = 1x1 cell array
    {1x1 sqw}

```

```

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



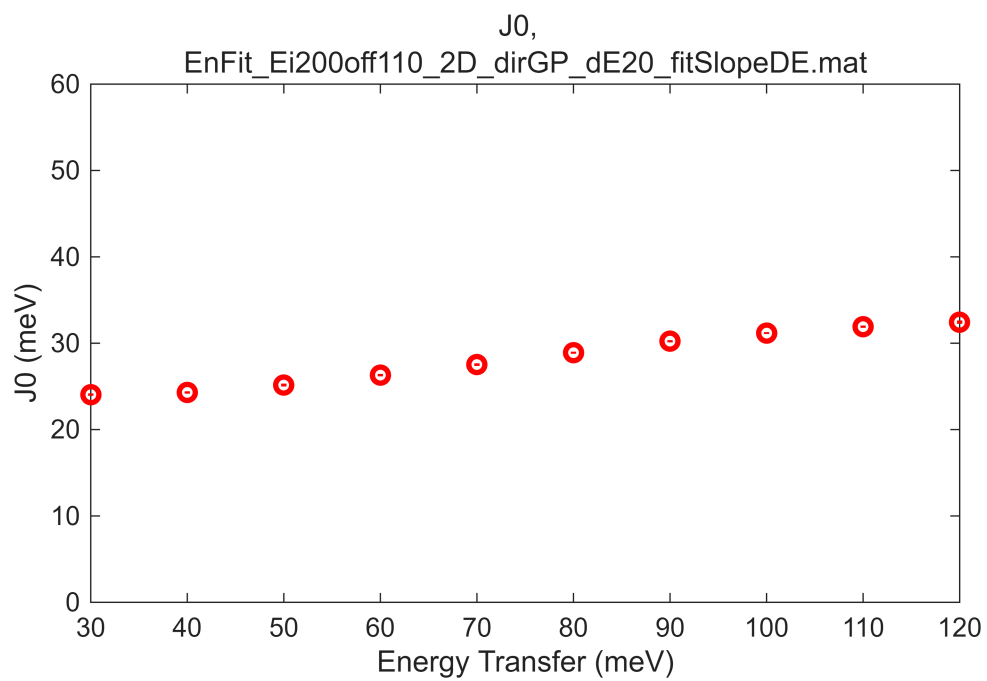
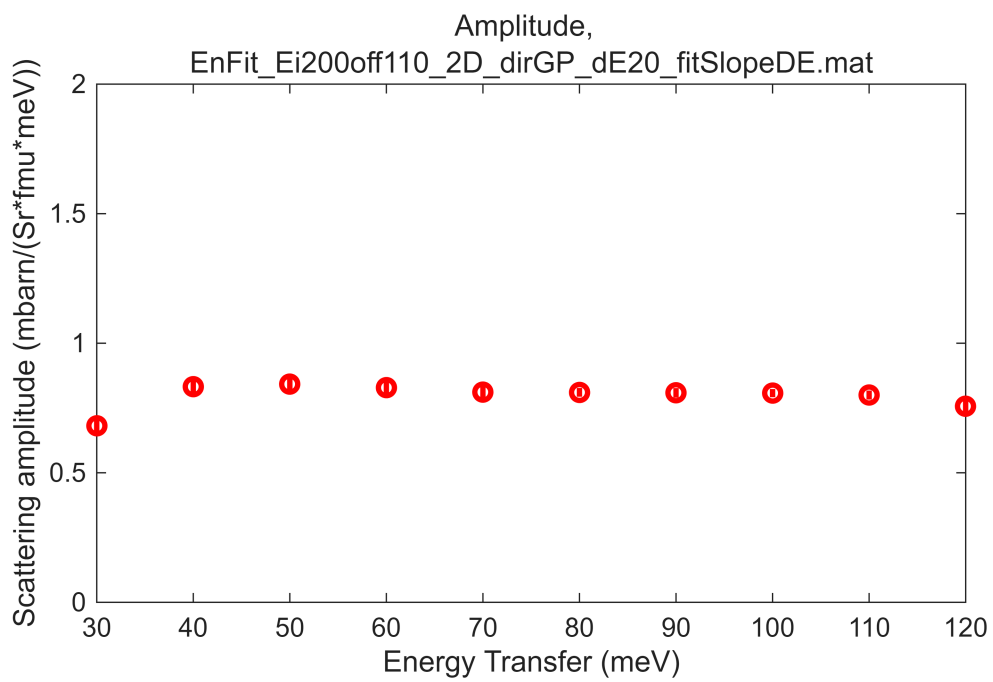
```
dir_name = "GP";

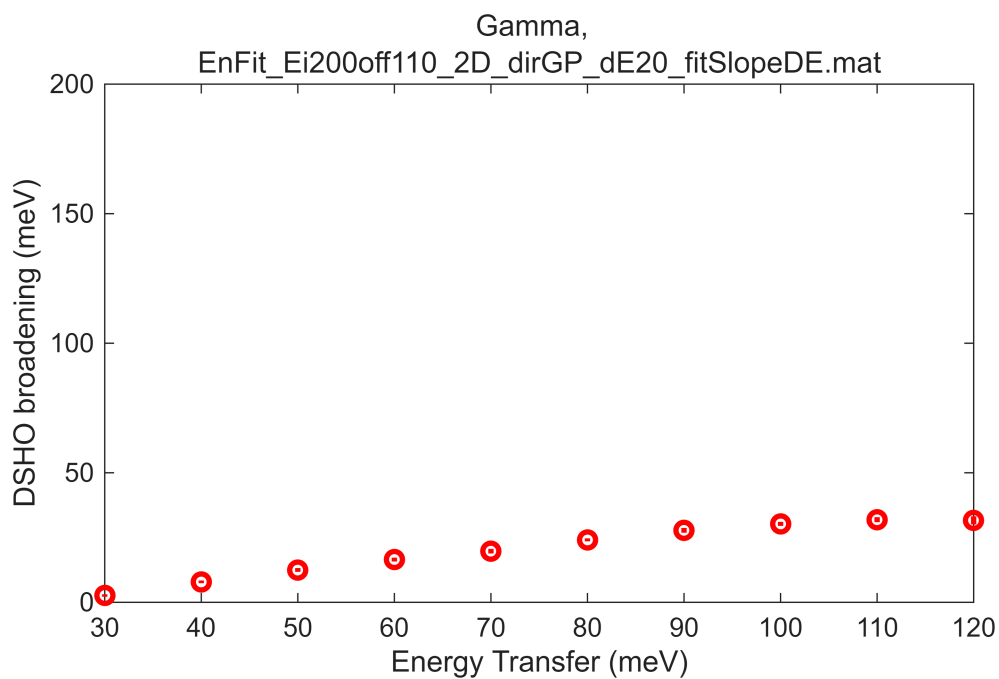
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning

cut_en = sel_info.en_range;
cut_dimensions = sel_info.cut_dim;

bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;
```

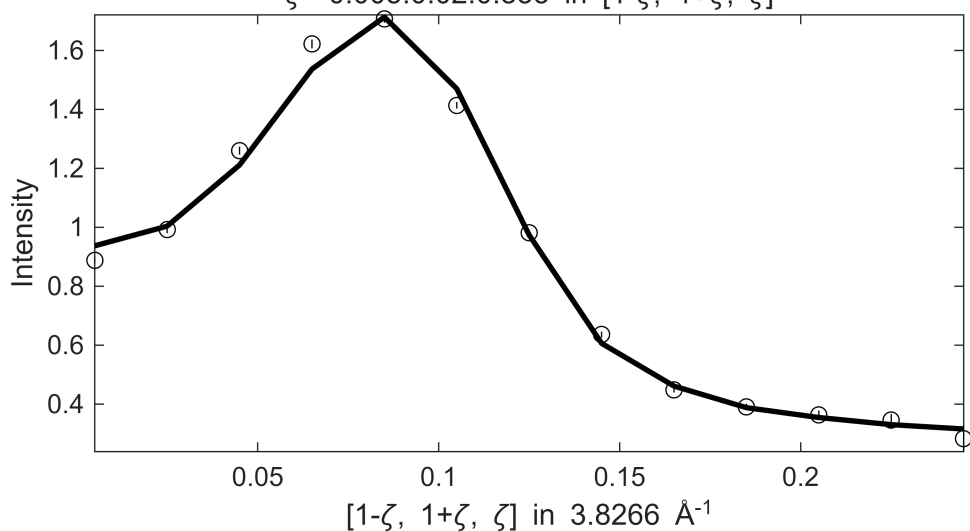
```
[fit_res_rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```



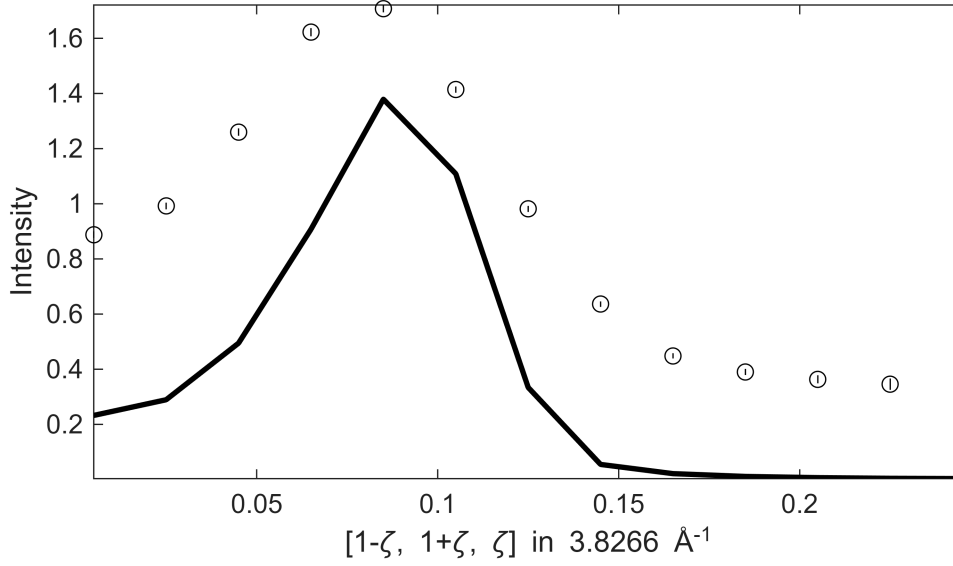


 **** En = 30 ± 10

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.68 \pm 0.012$; $J0= 24 \pm 0.14$; $\gamma=2.67378 \pm 0.0401818$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $20 \leq E$
 $\zeta=-0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

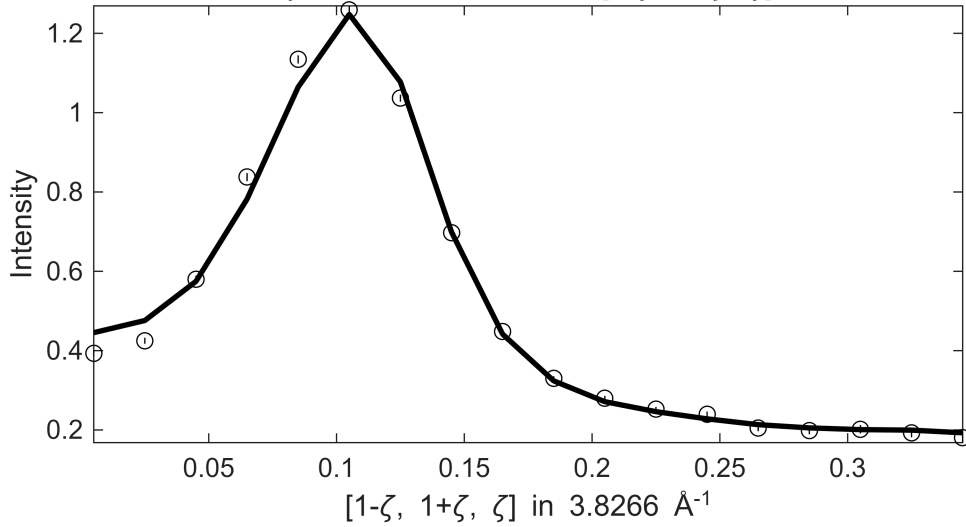


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $20 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



 **** En = 40 ± 10

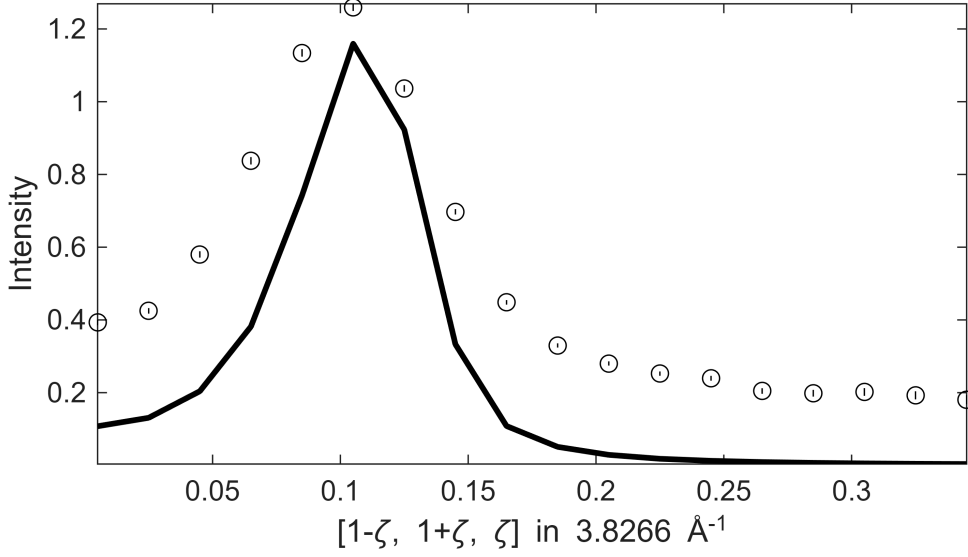
Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.83 \pm 0.012$; $J_0 = 24 \pm 0.12$; $\gamma = 7.87041 \pm 0.115553$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $30 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $30 \leq l$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



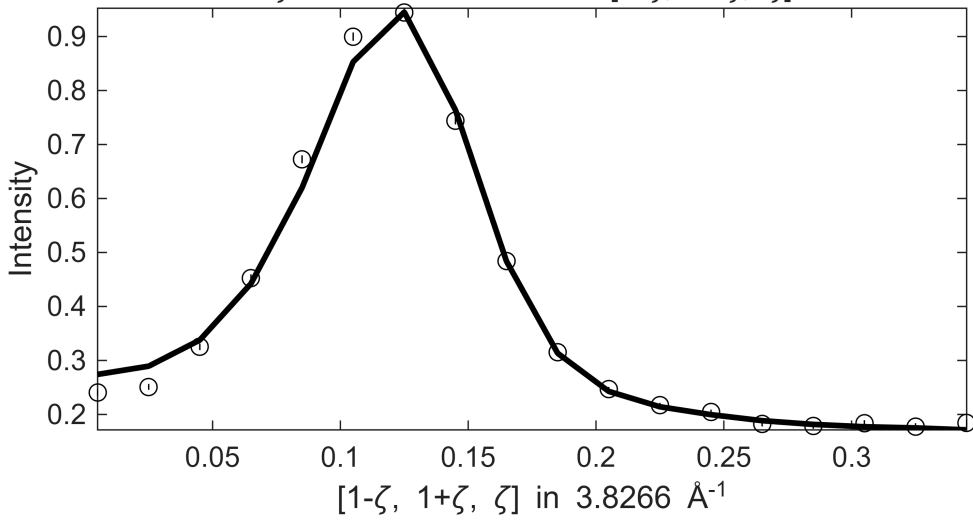
**** En = 50 ± 10

Fe_ei200_align_sym3, w2_200off110m.sqw

S= 0.84 ± 0.026 ; J0= 25 ± 0.17 ; gamma= 12.5038 ± 0.741231

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $40 \leq l$

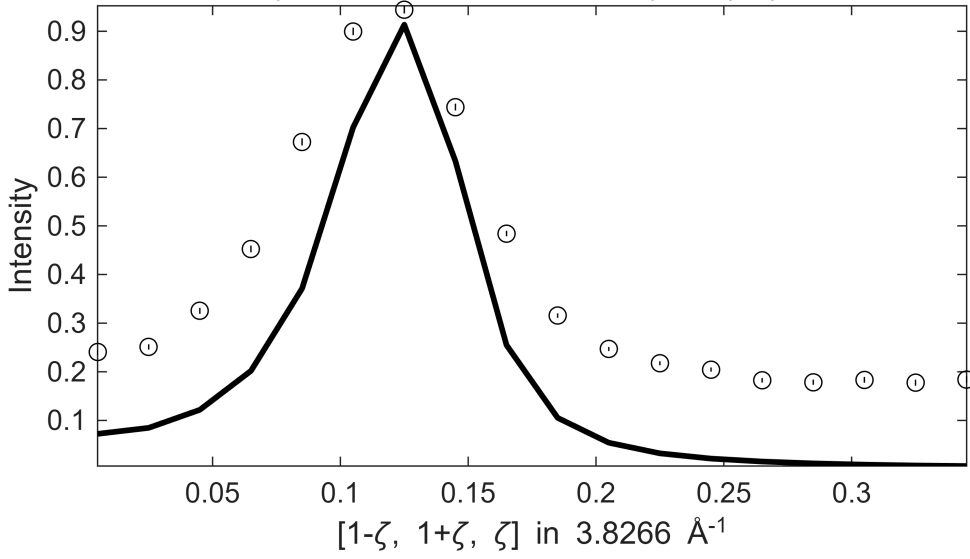
$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $40 \leq l$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



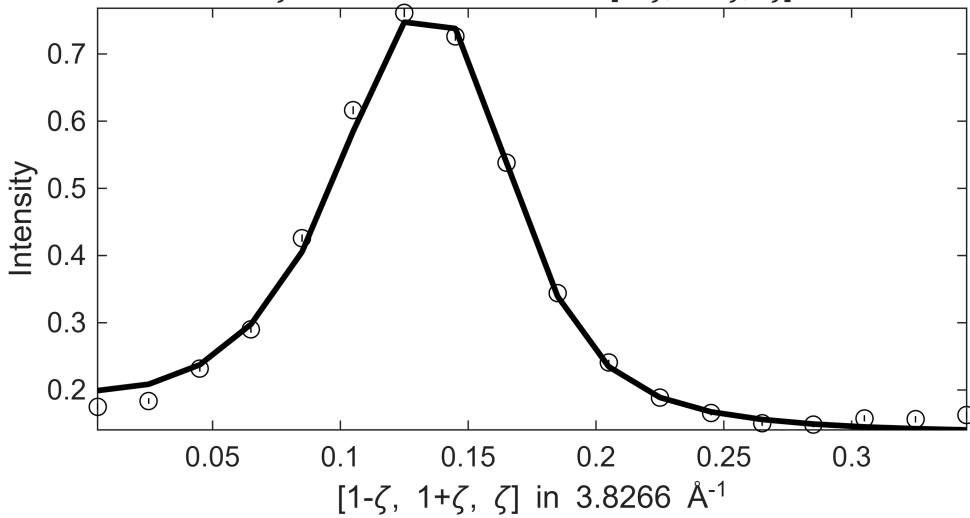
**** En = 60 ± 10

Fe_ei200_align_sym3, w2_200off110m.sqw

S= 0.84 ± 0.022 ; J0= 26 ± 0.15 ; gamma= 16.7044 ± 0.742116

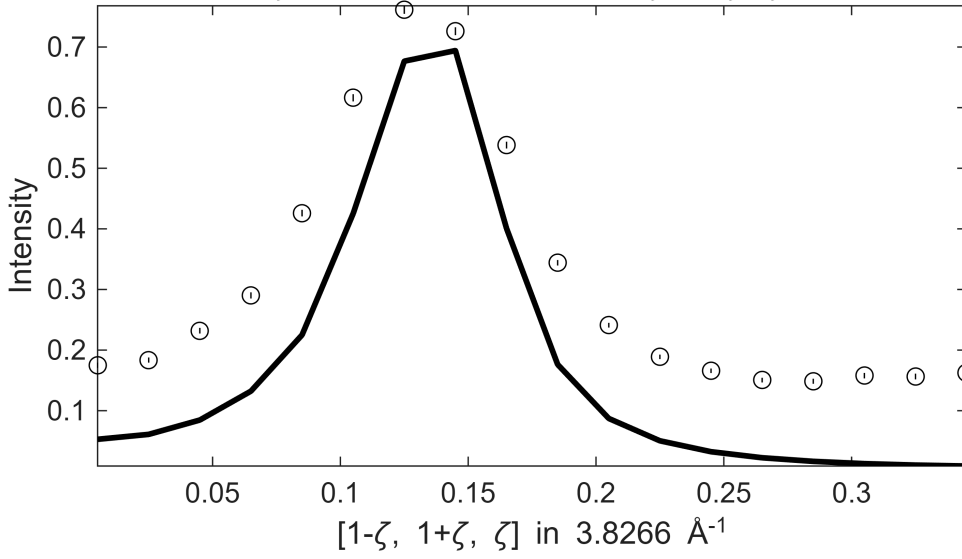
$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $50 \leq l$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $50 \leq l$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

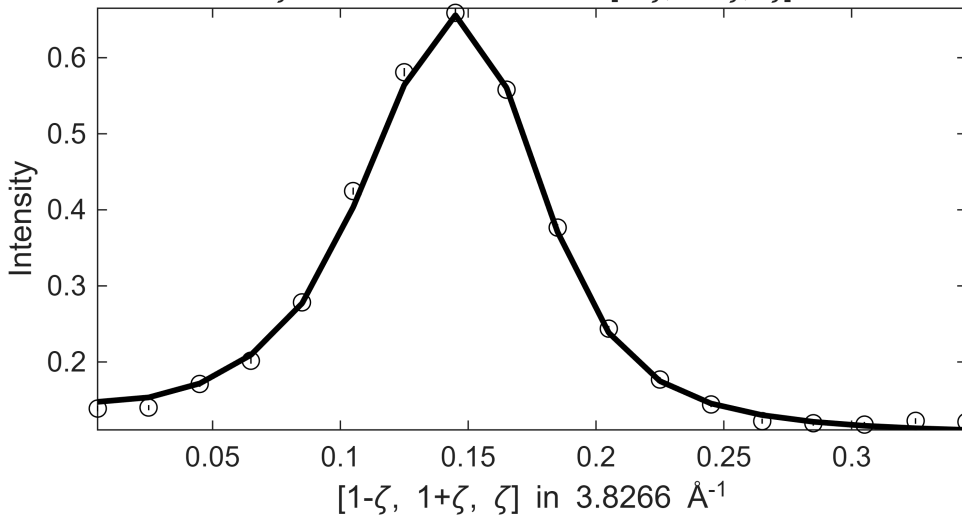


**** En = 70 ± 10

Fe_ei200_align_sym3, w2_200off110m.sqw

S= 0.81 ± 0.013 ; J0= 28 ± 0.12 ; gamma= 19.7548 ± 0.33481

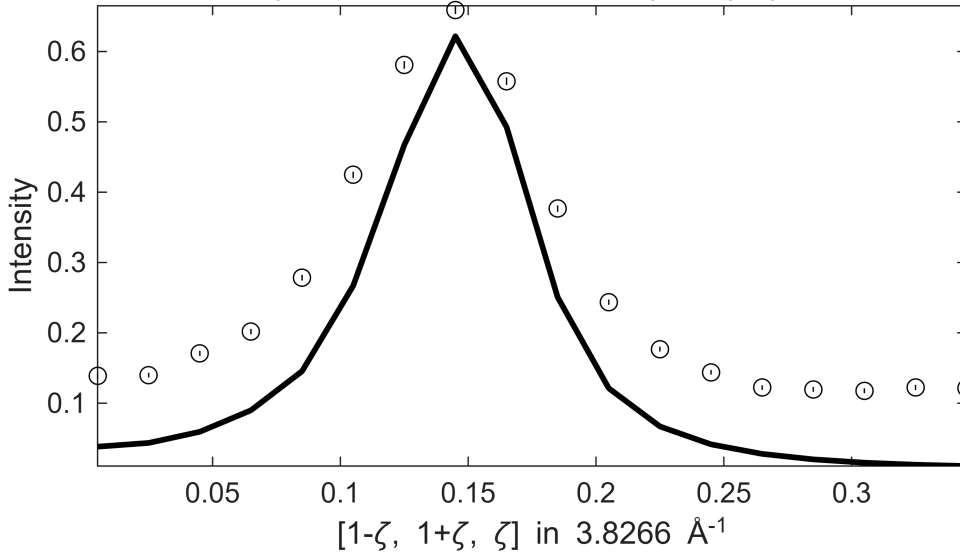
$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq l$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq l$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



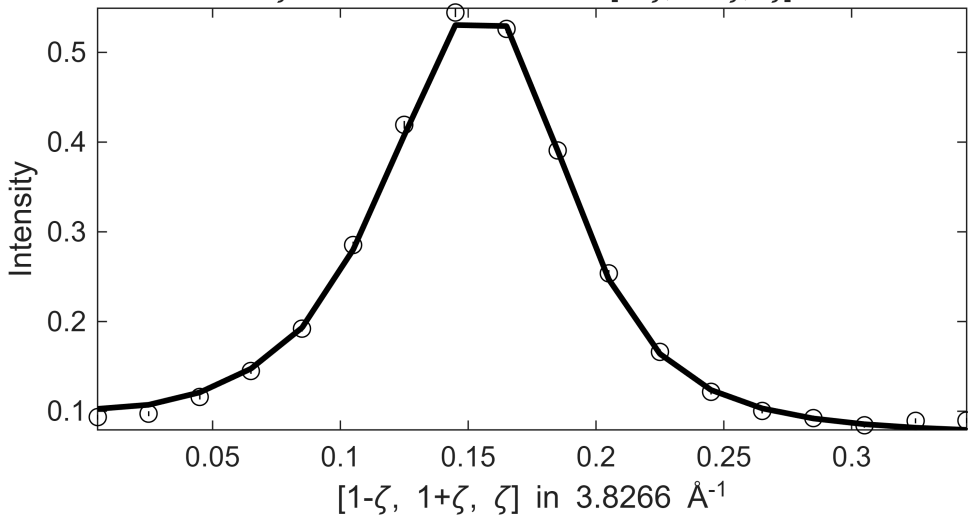
**** En = 80 ± 10

Fe_ei200_align_sym3, w2_200off110m.sqw

S= 0.81 ± 0.023 ; J0= 29 ± 0.16 ; gamma= 24.0856 ± 1.02667

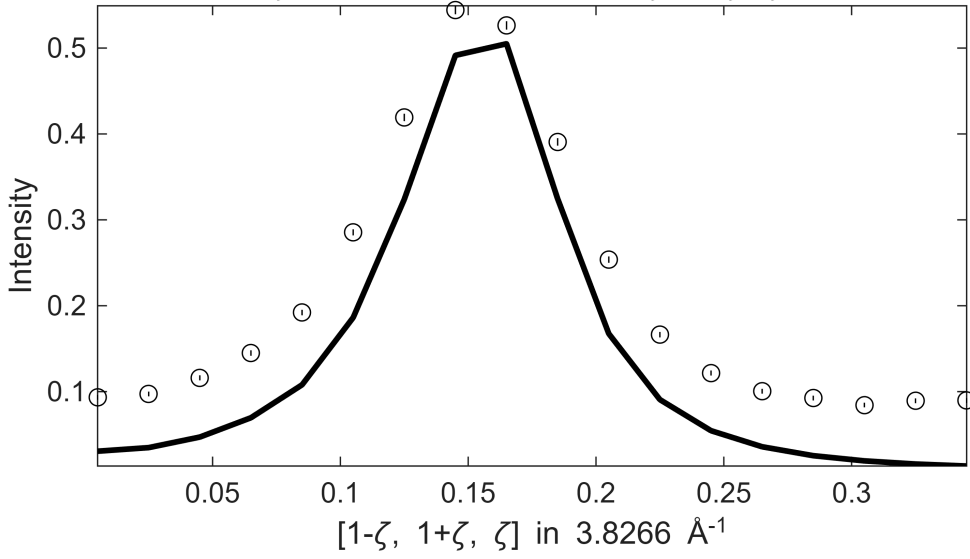
$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq l$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq l$
 $\zeta=-0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

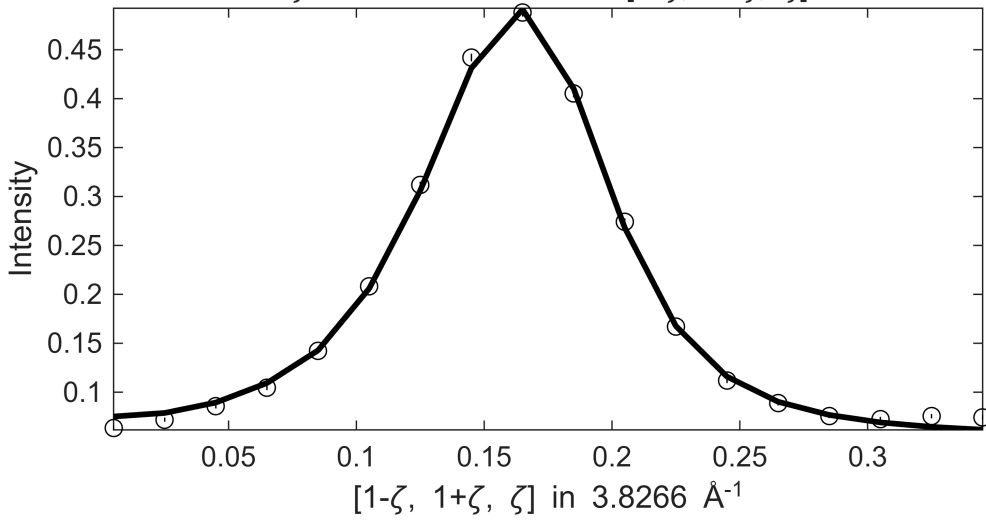


**** En = 90 ± 10

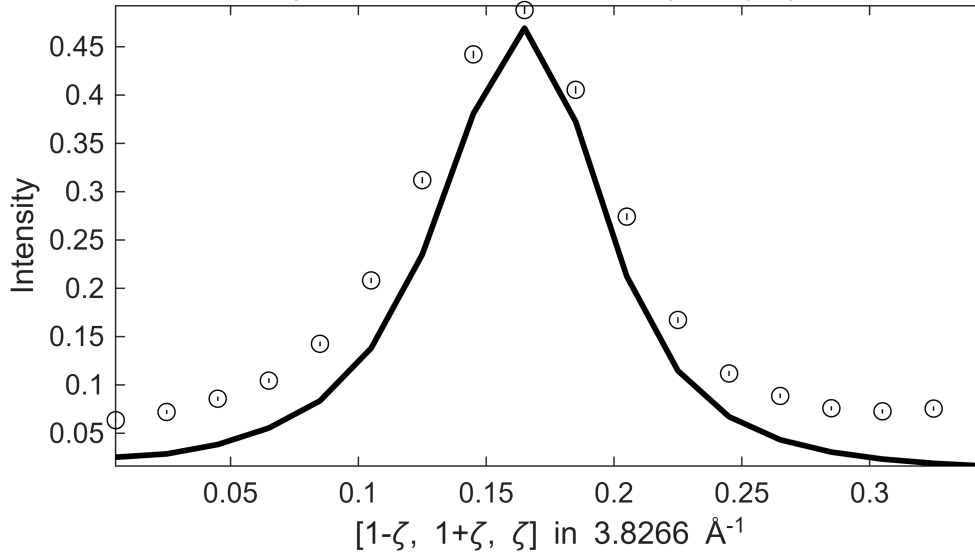
Fe_ei200_align_sym3, w2_200off110m.sqw

S= 0.81 ± 0.019 ; J0= 30 ± 0.13 ; gamma= 27.7813 ± 0.928091

$\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$
 $\zeta=-0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

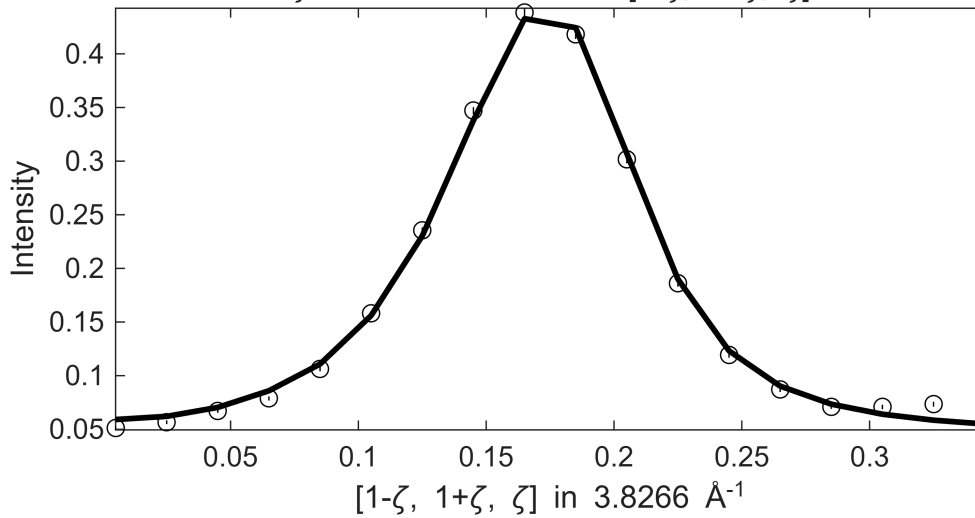


Fe_ei200_align_sym3, w2_200off110m.sqw
 $\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$
 $\zeta=-0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

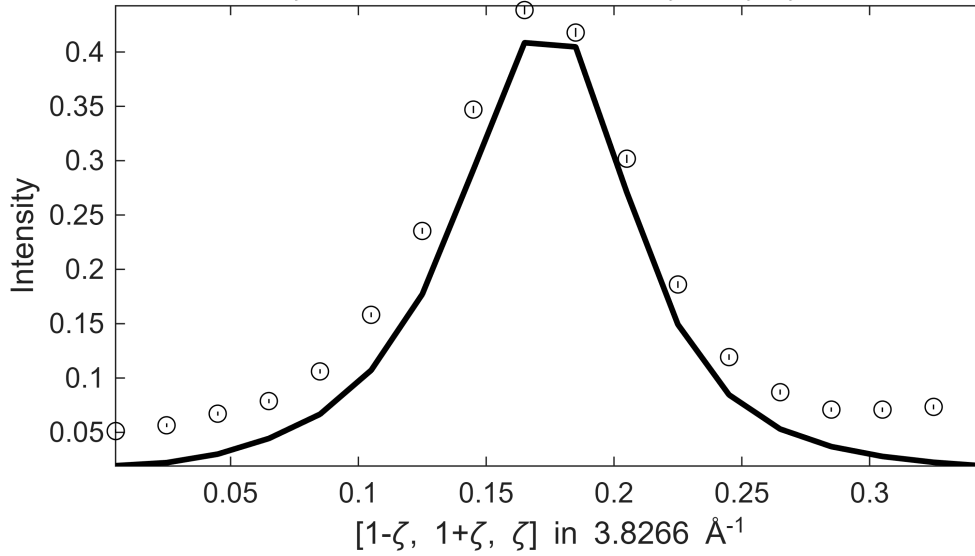


 **** En = 100±10

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.81\pm0.016$; $J_0= 31\pm0.11$; $\gamma=30.2673\pm0.778044$
 $\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$
 $\zeta=-0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw
 $\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

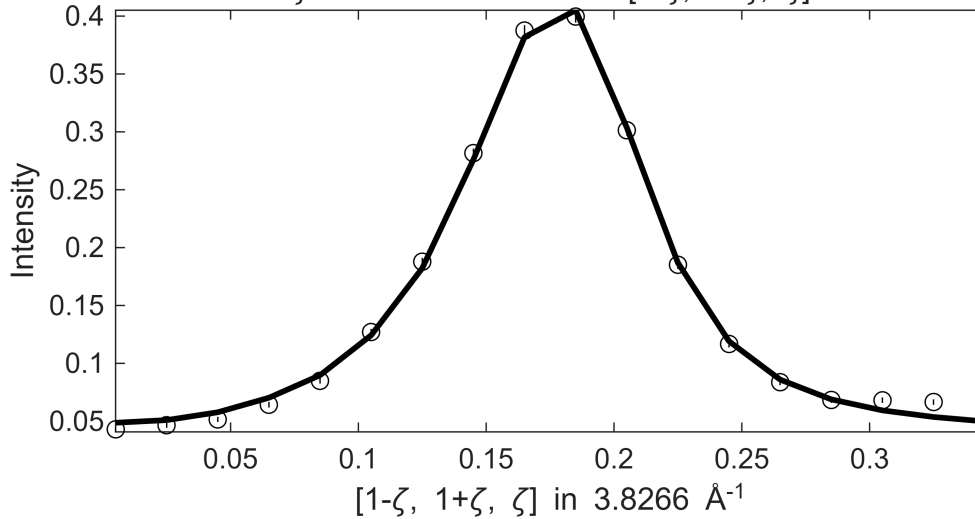


**** En = 110±10

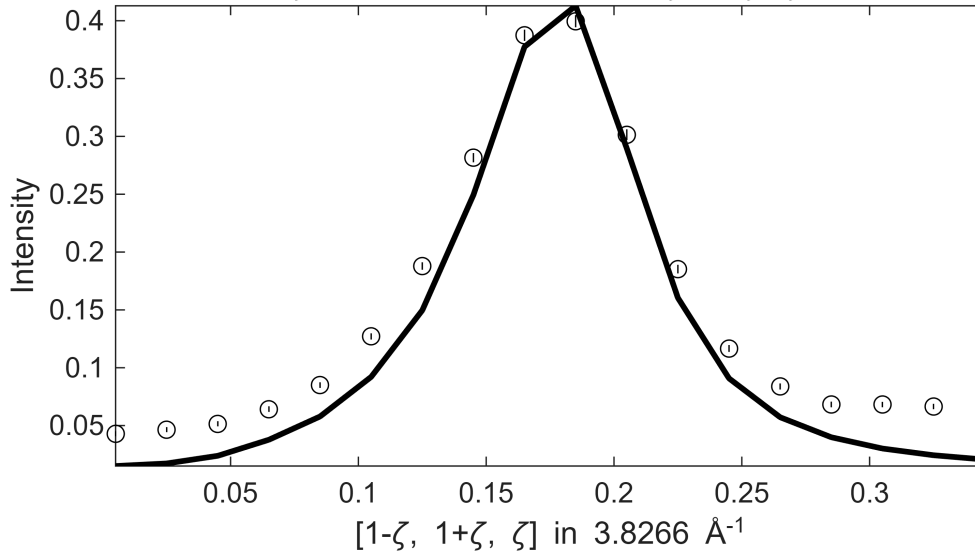
Warning: Dataset:1 points with undefined (infinite or NaN) data values have been removed from fit, which is 9.0909

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.5253

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S = 0.8 \pm 0.017$; $J_0 = 32 \pm 0.12$; $\gamma = 31.8841 \pm 0.906785$
 $\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw
 $\xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq t$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

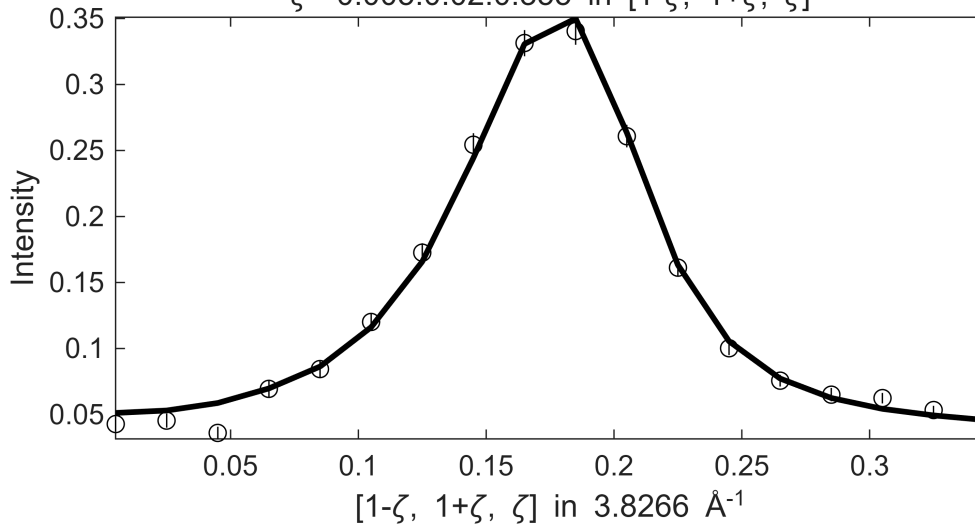


**** En = 120±10

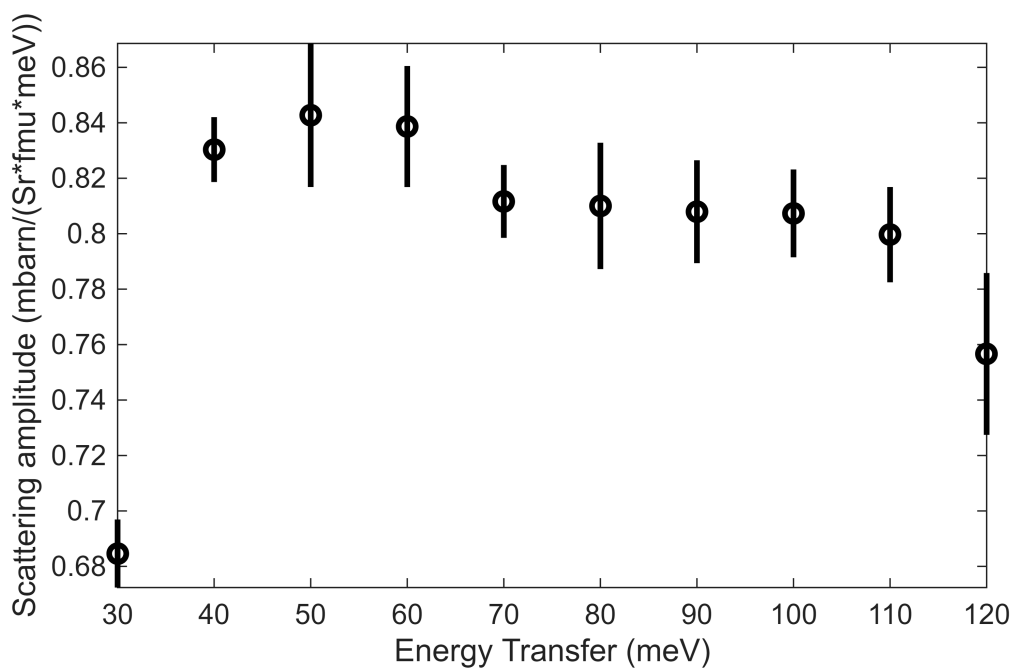
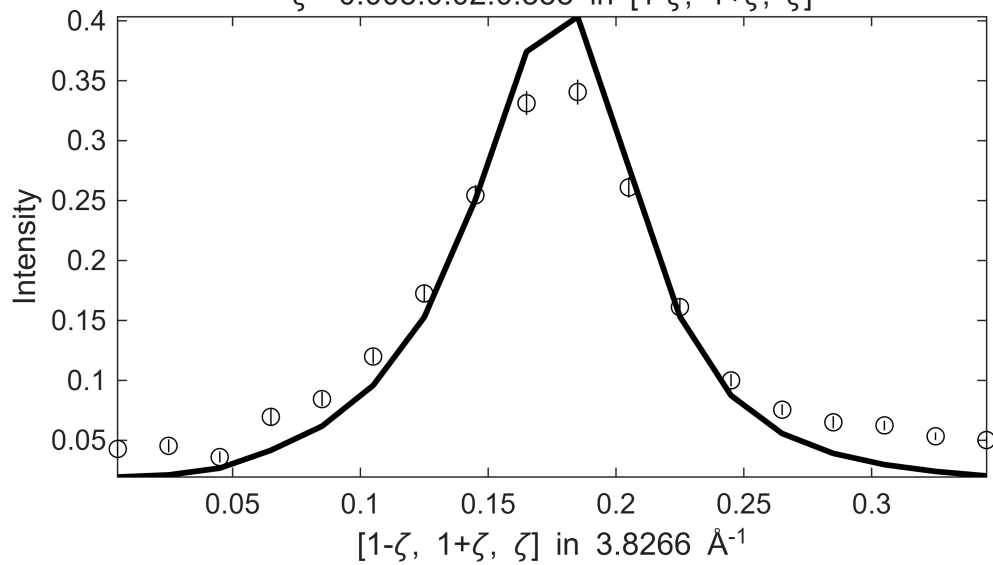
Warning: Dataset:1 points with undefined (infinite or NaN) data values have been removed from fit, which is 9.0909

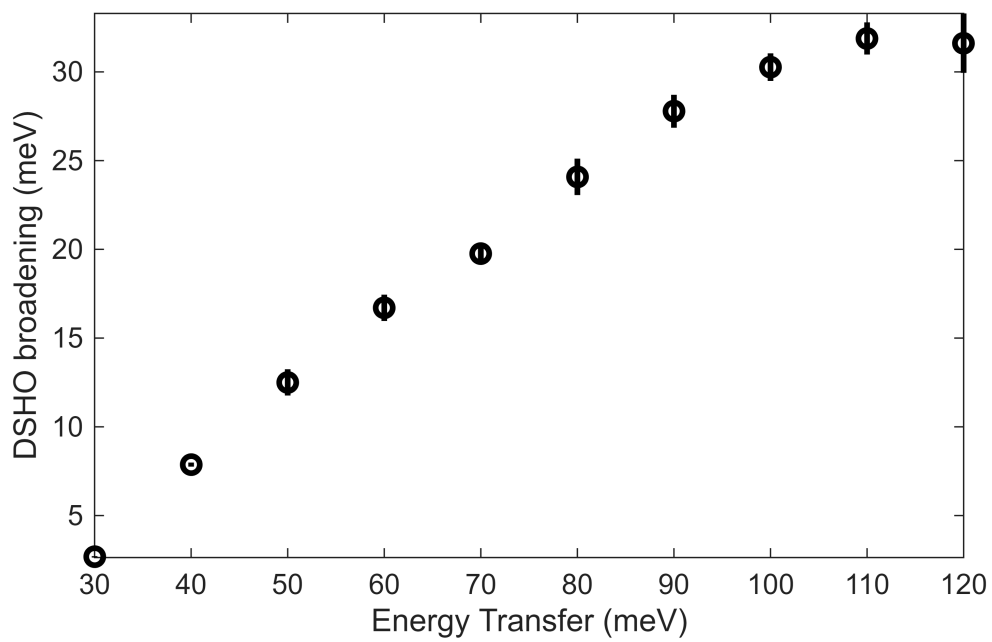
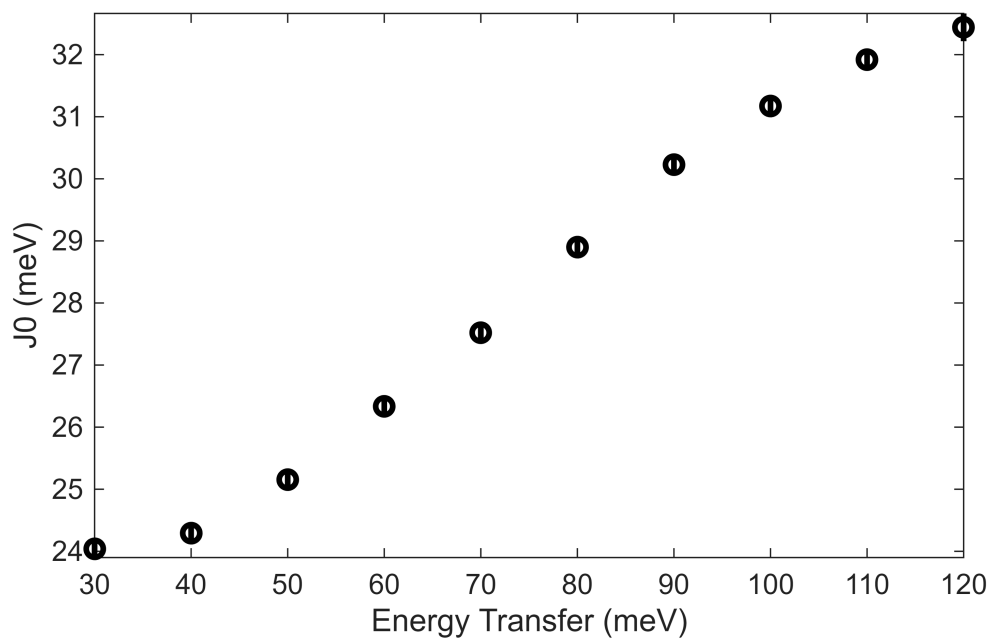
Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.5253

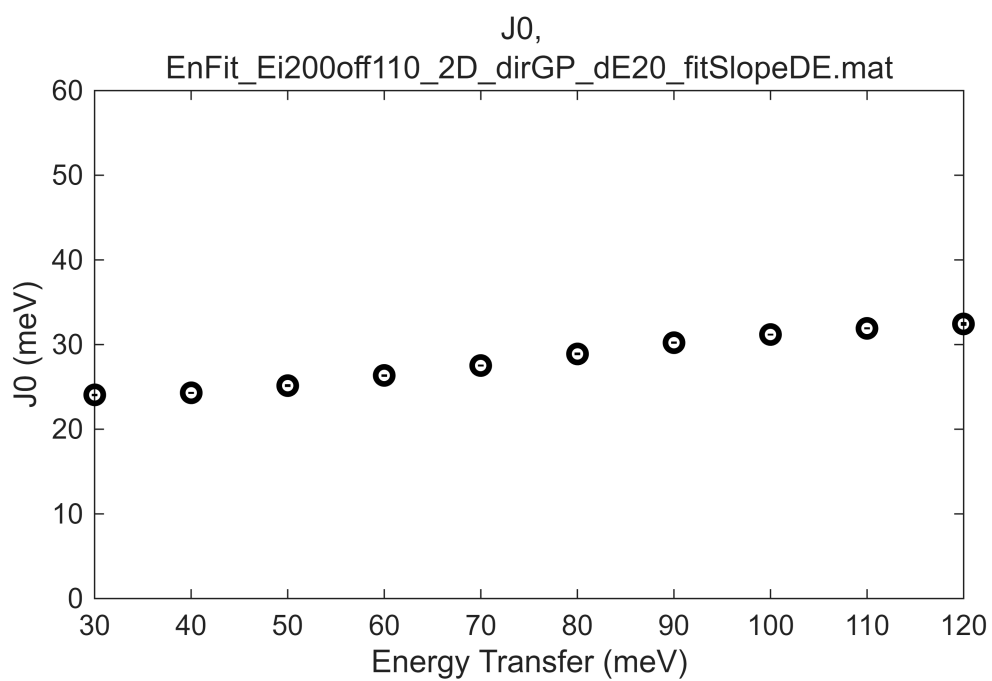
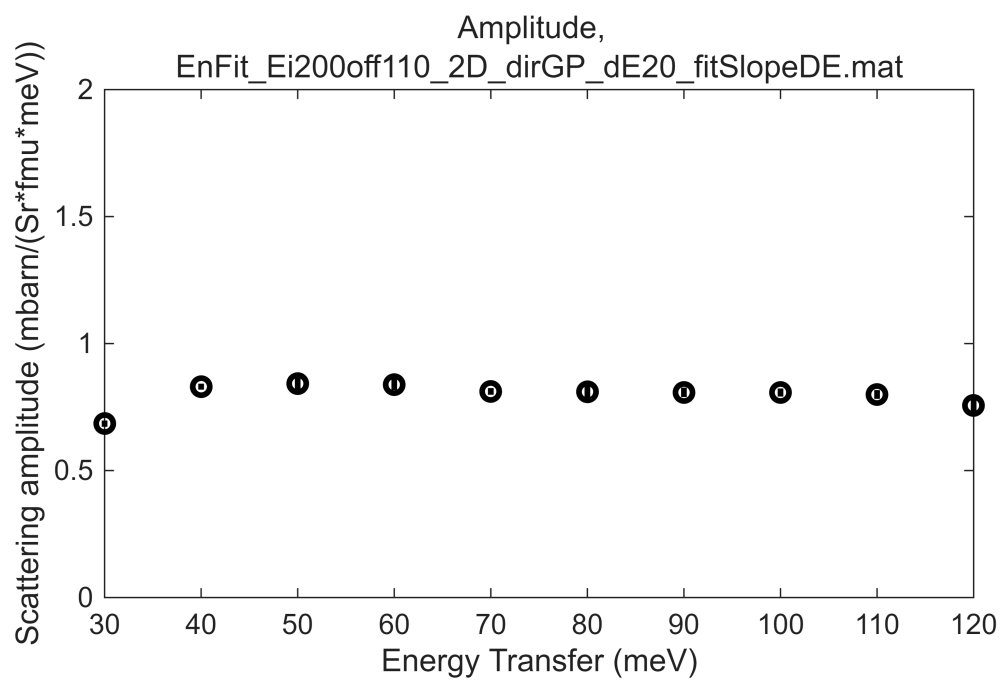
Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.76 \pm 0.029$; $J_0 = 32 \pm 0.22$; $\gamma = 31.6194 \pm 1.67316$
 $\xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $110 \leq t$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

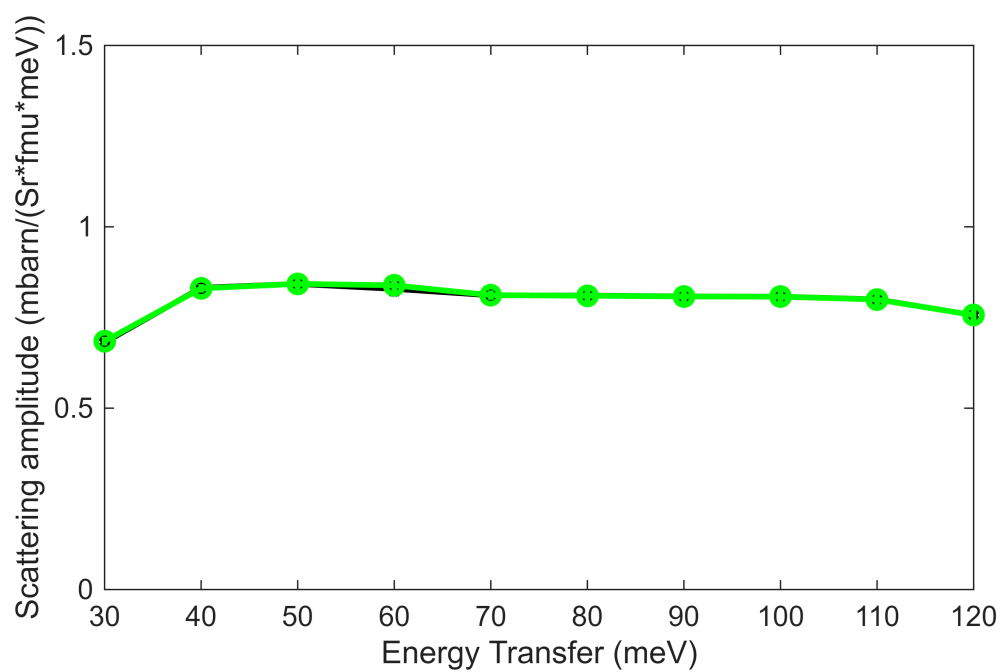
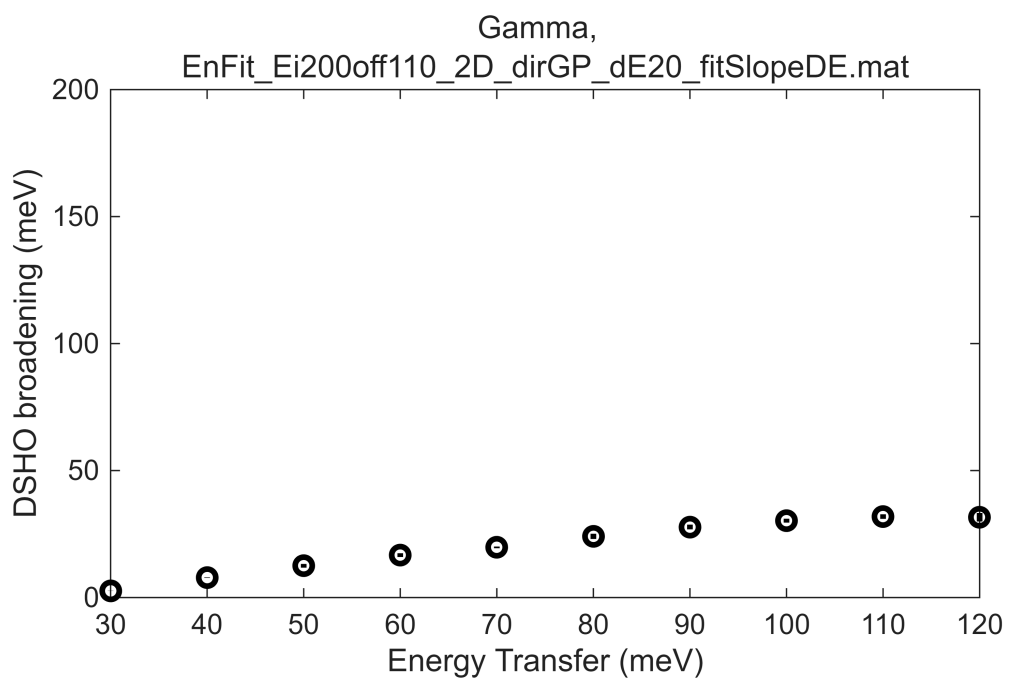


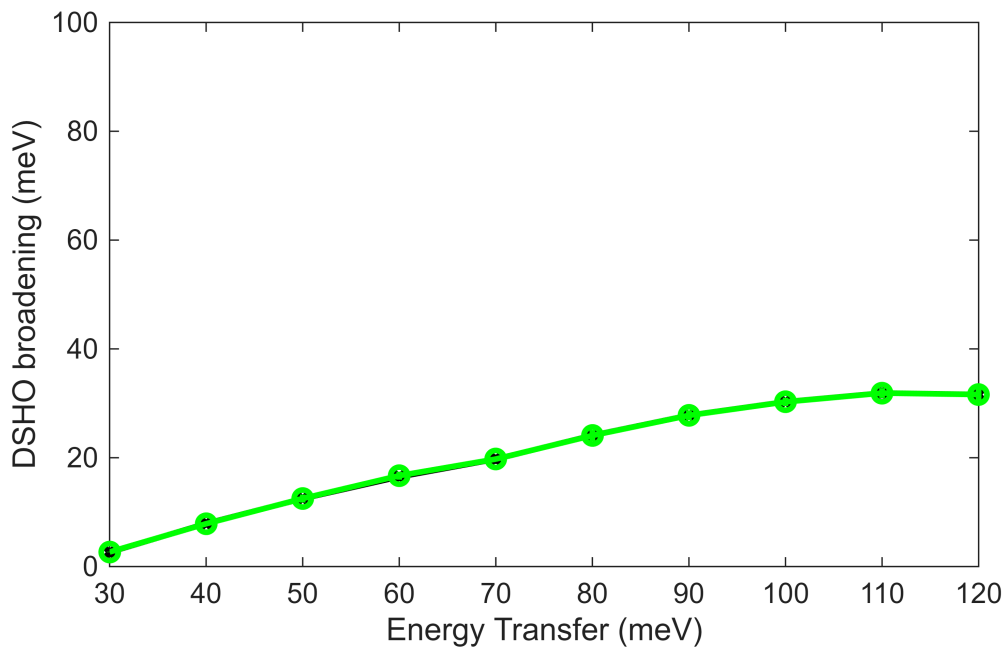
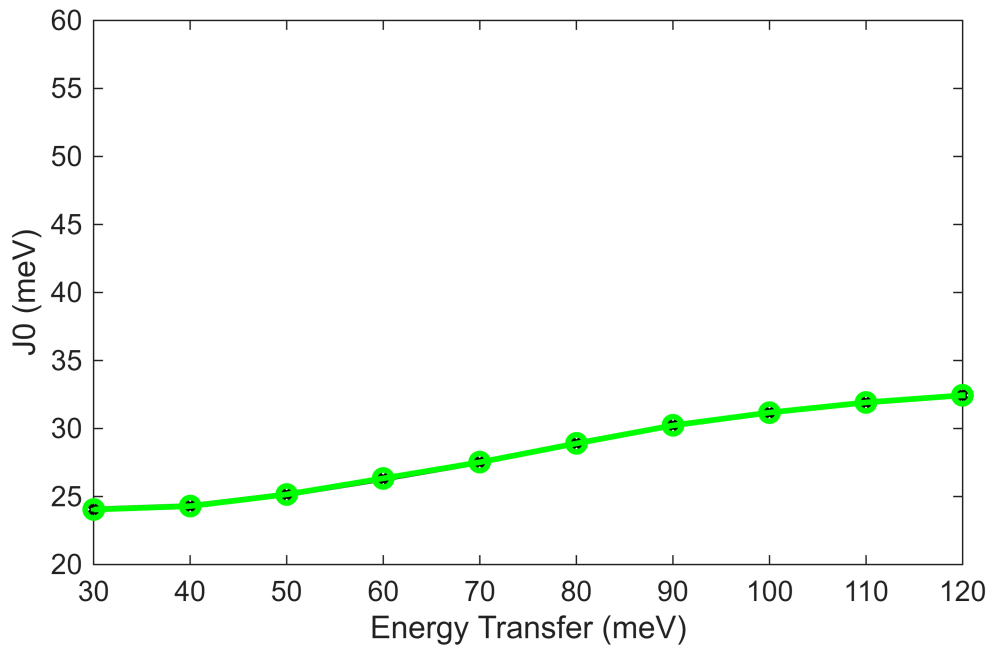
Fe_ei200_align_sym3, w2_200off110m.sqw
 $\leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $110 \leq \Gamma$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$











beep
beep
beep

```
bg_data = extract_bg_par(fit_res_rec.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure;
pd(bg_data(2));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
```

```
fit_resEi200 = fit_res_rec;  
save(res_name, "fit_resEi200")
```